

Class A Push-Pull: The Bad Cat Hot Cat 30 Power Amp



Ampbooks

Site: **Pull: The Bad Cat Hot Cat 30 Power Amp**

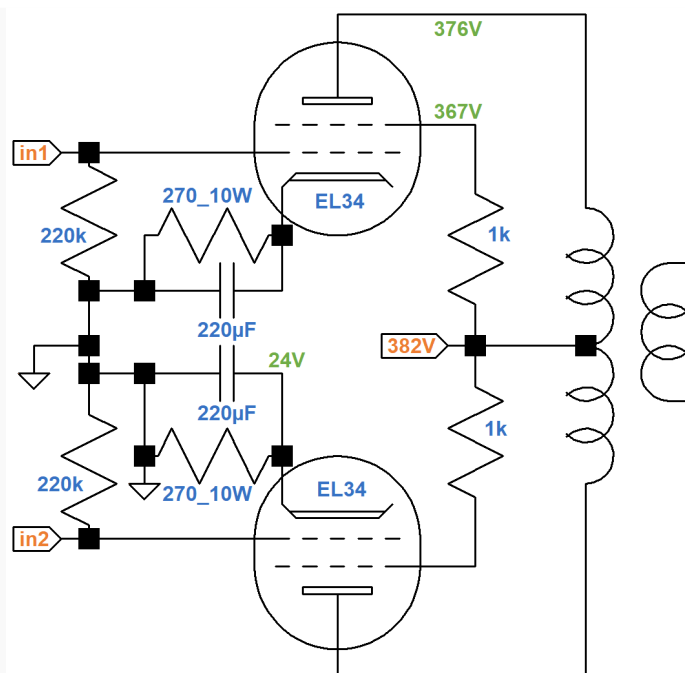
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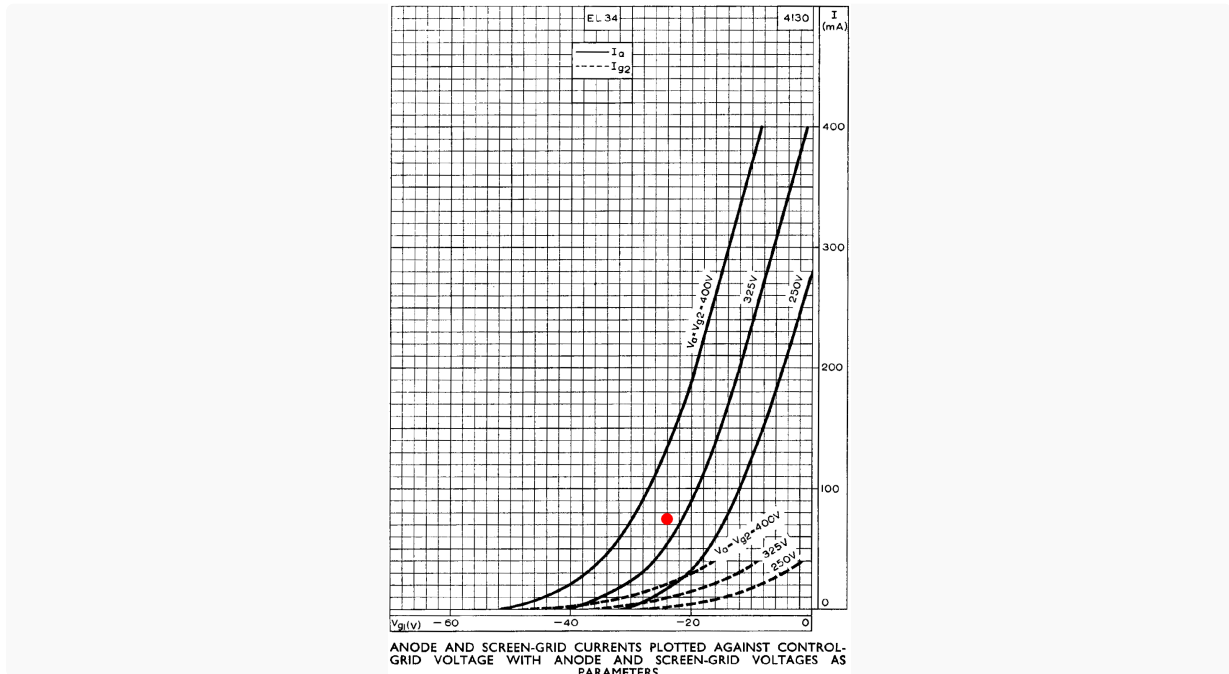
The Bad Cat Hot Cat 30 has an EL34 push-pull power amp biased for Class A operation, a rarity for guitar amps.

"Keen to build high-quality amps in the vein nominally referred to as 'Class A,' founder James Heidrich did his due diligence before starting up the company and, as he told me in the early 2000s, 'All roads seemed to lead to Mark [Sampson, founder of Matchless].' Sampson had just left Matchless at the time, was able to come on as Bad Cat's chief designer, and there we have it: no mere coincidence that cornerstone models such as the Black Cat and Cub looked extremely like the DC30 and Lightning, respectively, in almost every way - other than their use of a few modifications that Sampson might have liked to bring to the Matchless amps had he stayed." 1 -Dave Hunter

The Hot Cat design is cathode biased with a separate cathode resistor for each tube. There is no negative feedback from the output transformer secondary.



With a 367V screen and a 24V cathode, the screen-to-cathode voltage is 343V. An imagined 343V curve on the EL34 plate transfer characteristics intersects the vertical line for a -24V grid at the red dot shown here.



The Y axis indicates that the DC plate current is 75mA per tube. The plate-to-cathode voltage is $376V - 24V = 352V$. Idle plate dissipation is

$$(352V)(75mA) = 26W$$

This is only slightly higher than the 25W limit for an EL34. The key takeaway is Hot Cat's hot bias, approximately 100-percent plate dissipation, which enables it to get the most power from a Class A design. Class AB designs are typically biased at less than 85 percent, with 70 percent being a widely followed rule of thumb.²

Bad Cat measures a voltage drop of 15V across each 1k Ω screen resistor. According to Ohm's Law, the idle screen current is

$$15V / 1k\Omega = 15mA$$

The total cathode current is $75mA + 15mA = 90mA$. According to Ohm's Law, this creates a voltage drop across a 270 Ω cathode resistor equal to

$$(90mA)(270\Omega) = 24.3V$$

This matches Bad Cat's measured value.

We get another perspective on the design when we consider that the plate-to-cathode voltage is only 9V higher than the screen-to-cathode voltage. This is close enough that we can use the triode-connected curves and assume the triode's plate is at 343V. The red dot marks the DC operating point for 343V plate-to-cathode and 90mA "plate" current.

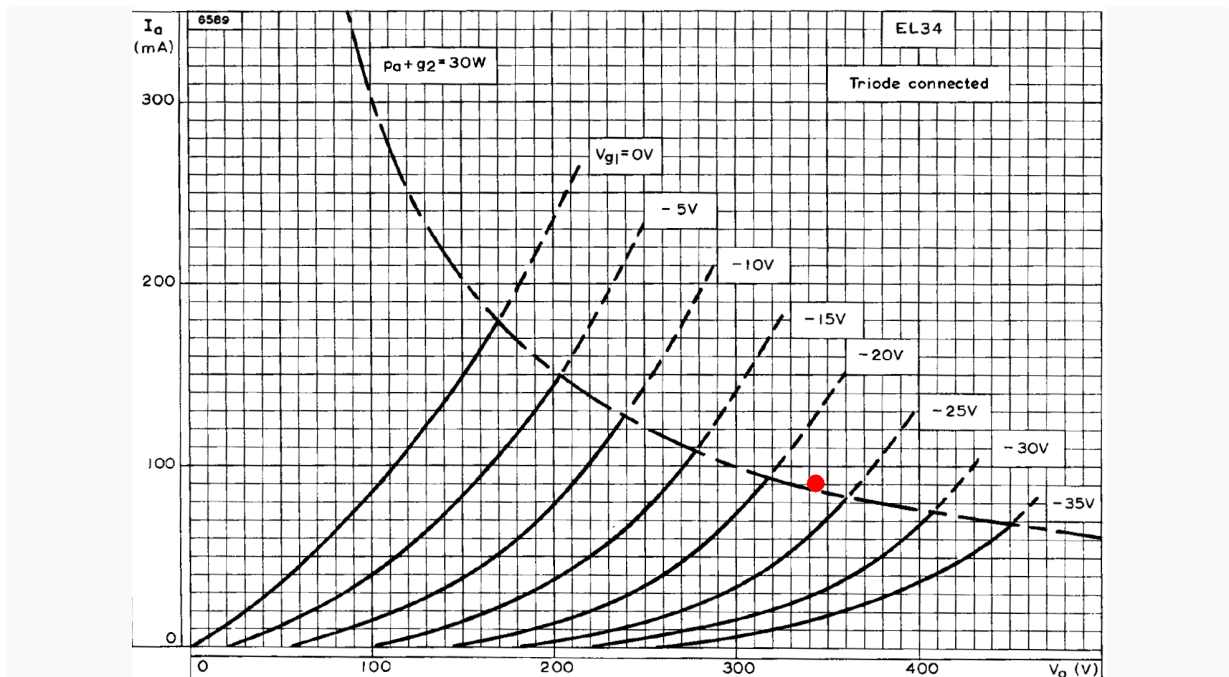
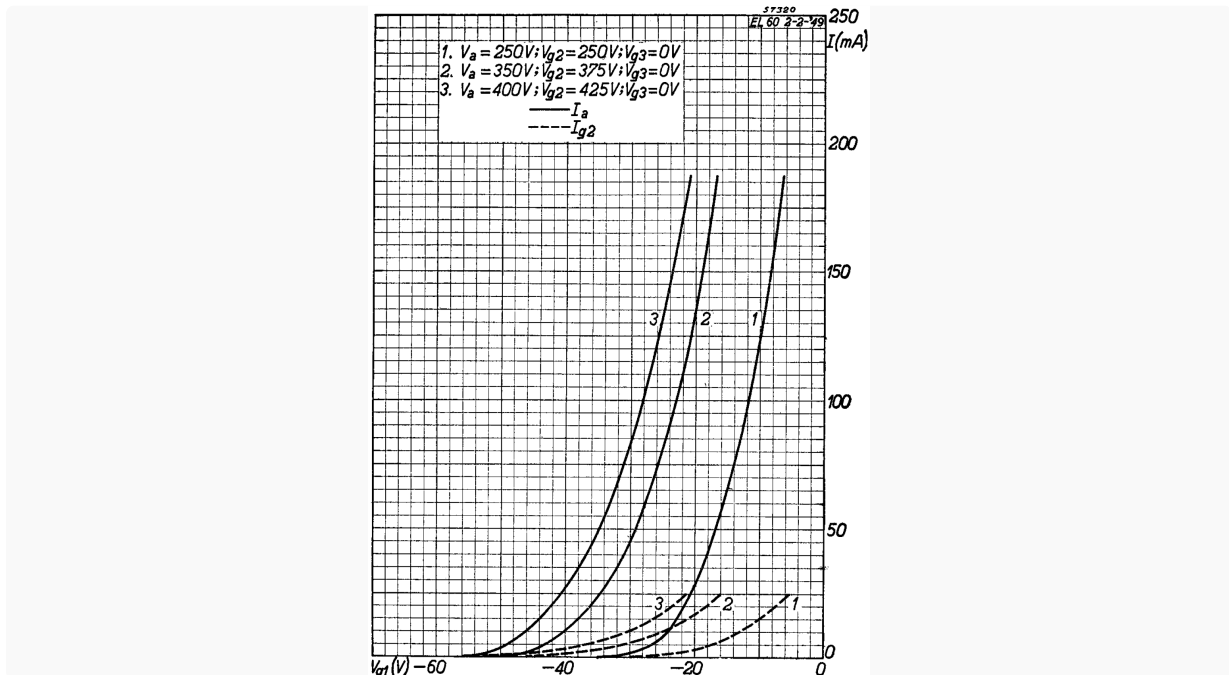


Plate and screen dissipation combined is just slightly above the 30W limit shown by the dashed curve.

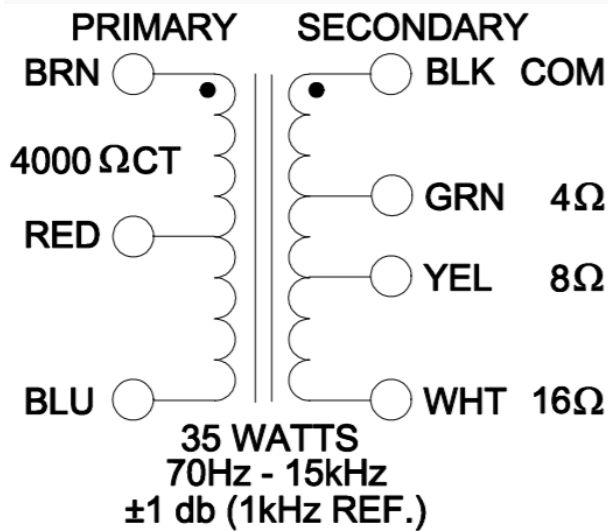
For a traditional Class AB design, each tube is in cutoff for a substantial portion of each cycle when the power amp is driven to full power. If the DC grid bias is -24V, for example, then the cutoff grid voltage is less negative than -48V. Let's say it is -40V. When one grid swings to -8V, the other grid swings to -40V and its tube transitions into cutoff. As the voltage at the first grid increases further toward 0V, the other tube remains in cutoff as its grid voltage decreases toward -48V. For the portion of each cycle that puts the grid voltage between -40V and -48V, the tube is in cutoff and the other tube is the sole driver of plate current through the output transformer.

For comparison, let's examine what happens for the Hot Cat power amp when a grid swings to -48V. Here is another set of EL34 plate transfer characteristics.



Solid Curve 2 is for a 375V screen and a 350V plate. It shows a cutoff grid voltage of about -48V. When a Hot Cat grid reaches its full-power limit of -48V, plate voltage swings to its positive limit, which is much higher than 350V. Screen current is minimal, as shown by dashed Curve 2, so the screen voltage increases to approximately the screen supply voltage: 382V. Both of these factors create a cutoff grid voltage that is more negative than the -48V shown by solid Curve 2. At full power the tubes never reach cutoff. Instead they drive plate current throughout each cycle, representing Class A behavior.³

The schematic does not show the primary impedance of the TDS 1163 output transformer, but it may be 4kΩ plate-to-plate. The Hot Cat has an advertised power rating of 30 watts. Let's examine the results if we use a Hammond 1760J, which has a 4kΩ plate and a 35W power rating.

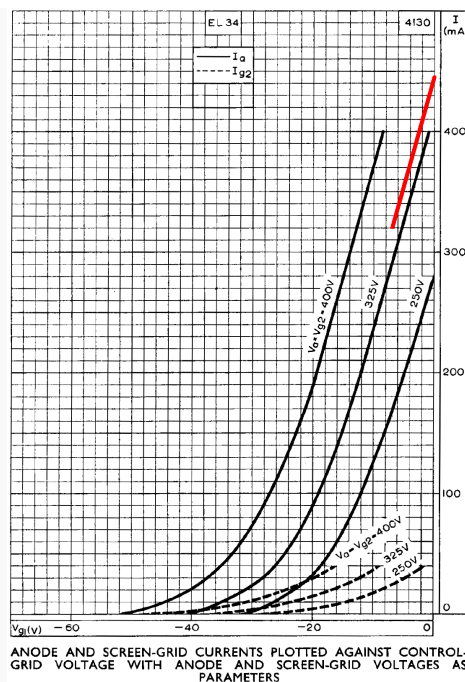


ELECTRICAL DATA:	
Turns ratio:	31.11 : 1
Inductance	@240V, 60Hz => 90.95H (Open circuit)
DCR	@ 20C, BRN - BLU = 185.0 Ω $\pm 20\%$
	@ 20C, BLK - GRN = 0.40 Ω $\pm 20\%$
FREQ. RESP	@ 35W(374.1V), 70Hz - 15kHz, BRN - BLU ± 1 db(1kHz REF.)
Pri. Impedance	4000 Ω CT (374.1V)
Sec. Impedance	4/8/16 Ω (11.83/16.3/23.66V)
Power	35 WATTS

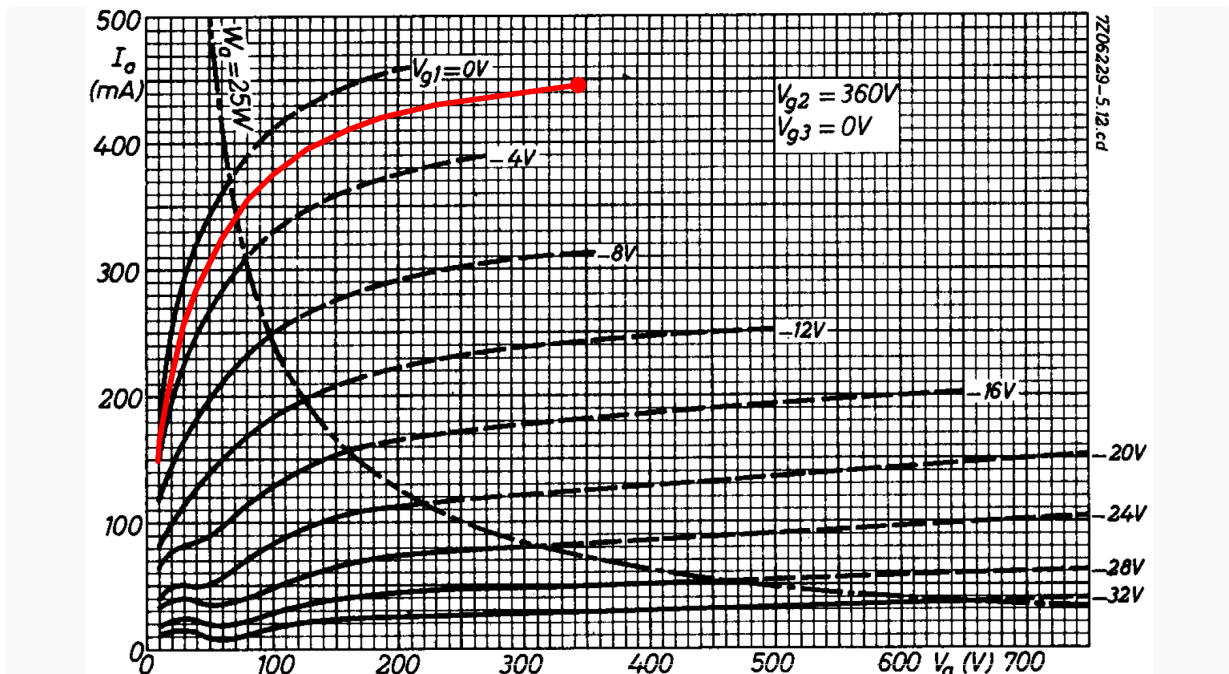
The data sheet indicates that the plate-to-plate DC winding resistance is 185 Ω , i.e. 93 Ω per phase. According to Ohm's Law, for an idle DC plate current of 75mA, the DC voltage drop across the primary is

$$(75\text{mA})(93\Omega) = 7\text{V}$$

Bad Cat measures a 6V drop for its transformer. This is far from conclusive evidence for a 4k Ω primary impedance, but it is a good start. Let's take another look at the transfer characteristics.



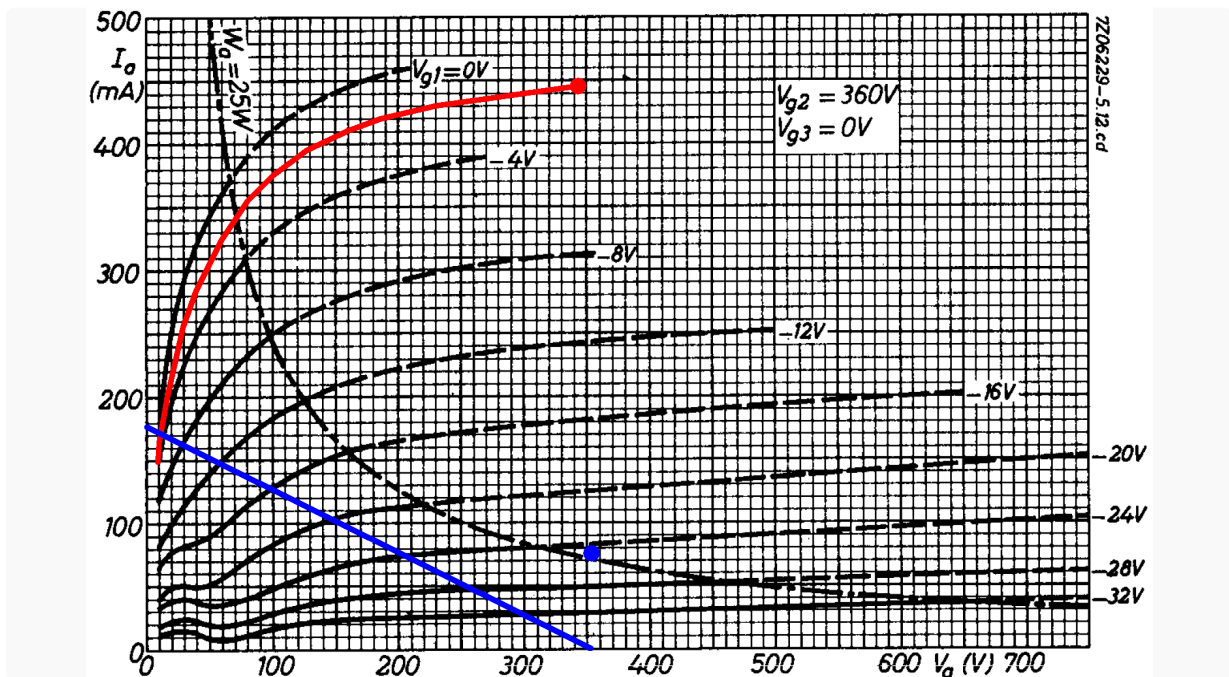
When the screen voltage is 343V, the plate voltage is 343V, and the grid voltage is 0V, the plate current is 445mA, as shown by the upper right endpoint of the red curve. This point can serve as an anchor point for a 0V curve on the plate characteristics for a 343V screen, as shown here by the red dot.



Each tube sees half the 4k Ω plate-to-plate impedance: 2k Ω . If the plate voltage swings from 352V to zero, the plate current increases from 0mA to

$$352V / 2k\Omega = 176mA$$

This defines the blue load line, which extends from 0V, 176mA to 352V, 0mA.



If this were pure Class B, the load line would represent the relationship between plate current and plate voltage for one tube. This is Class A, however, so the load line only represents *net* plate current. At idle when both grids are at -24V, for example, the plate current for each tube is 75mA in opposite directions through the output transformer primary. The net plate current is zero, as indicated by the load line. When the grid voltage increases from -24V to 0V, the actual relationship between plate current and plate voltage for one tube is closer to the maximum 25W dashed curve. For a -24V grid and a 352V plate, the plate current is 75mA, as shown by the blue dot. Plate current and voltage then trace a path above the blue load line all the way to the 0V grid curve, as required for Class A. Plate dissipation at full power is therefore substantially greater than the load line would seem to indicate. At both idle and full power, this cat is hot!

The load line intersects the red 0V curve at 12V, 172mA. Full power is therefore in the neighborhood of

$$(352V - 12V)(172mA) / 2 = 29W$$

This is close to the Hot Cat's 30W advertised rating. When a pair of EL34s are operating Class AB we expect to see at least 40 watts. Bad Cat thus sacrifices at least 10 watts to achieve its goal of Class A, which really shines through when driven by the clean channel.

"The Hot Cat's clean channel gives you a single tone control - a brilliance knob - but the basic sound is absolutely gorgeous. I strummed an open C chord and was immediately inspired by a stunning airy timbre. Upper-mid and high-end tones are rich and detailed, and no matter how loud I cranked up the amp, the overall sound was smooth and musical. The clean tone definitely cuts through a band mix - and diming the channel brings in some growl and sustain - but it never gets shrill or ragged." 4 -Darrin Fox

References

1Dave Hunter, "Dave's Corner: Boutique Amp Guide Part IV," **Reverb**, August 19, 2015, Available at <https://reverb.com/news/daves-corner-boutique-amp-guide-part-iv> (Retrieved June 30, 2020)

2Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018) p. 117.

3Richard Kuehnel, [Guitar Amplifier Electronics: Basic Theory](#), (Seattle: Amp Books, 2018) pp. 115-117.

4Darrin Fox, "Big Bottle Boogie," **Guitar Player**, August 2002, pp. 145-147.